

CMSE 381

Honors Option

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Research Questions

- **RQ1:** Do neural population responses in the AM face patch naturally organize by head orientation or by individual identity, revealing the dominant organizing principle of face representations in this region?
- **RQ2:** Do neural firing patterns in the AM face patch contain sufficient information to decode head orientation from temporal activity profiles, and which time windows are most critical for orientation discrimination?

(Explored through k-means clustering)

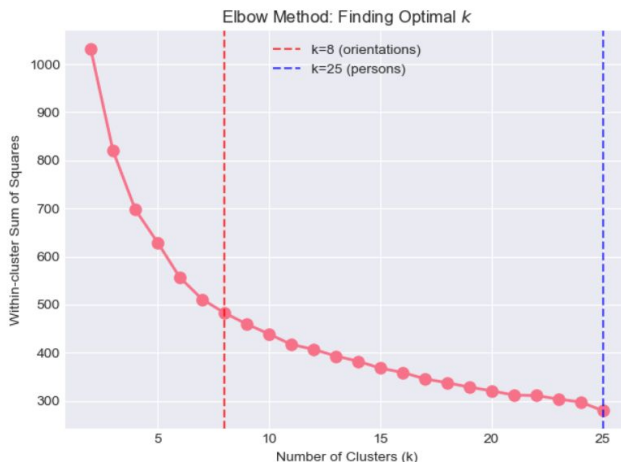
(Explored through decision tree classification)



[https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_\(Macaque\)_Mother_Grooms_Her_Young.jpg](https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_(Macaque)_Mother_Grooms_Her_Young.jpg)

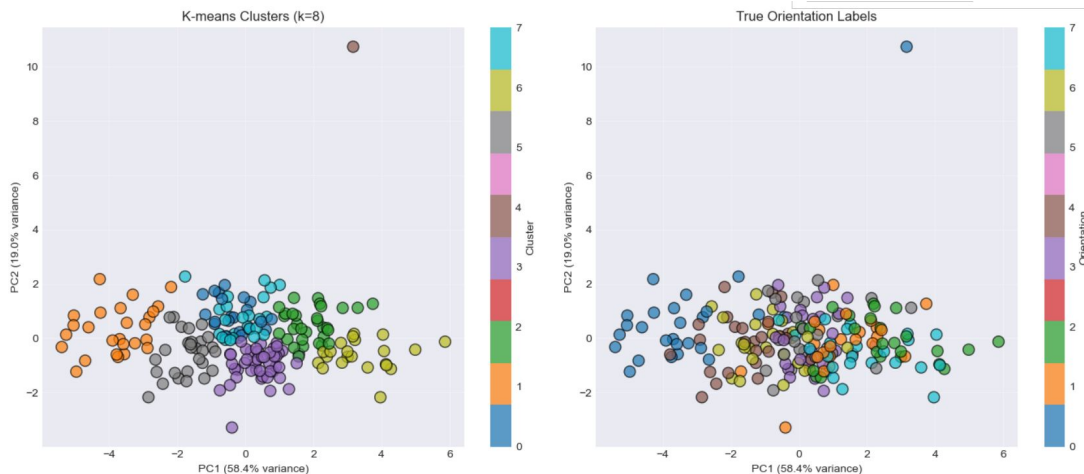
Methods Overview I

- Preprocessing: create a 200x8 feature matrix with mean firing rate for every unique combination of 25 classes and 8 head position
- Conduct PCA to make visualization of data points easier
- Compared k-means clustering with the original/true class labels PC1 v PC2



Key Findings I

- Elbow shows optimal k is before 8, but cannot be beyond 8.
- Visually, we can see clusters are possible, and clustering is slightly similar to the true orientation labels.
- This needs confirmation through actual classification, though.

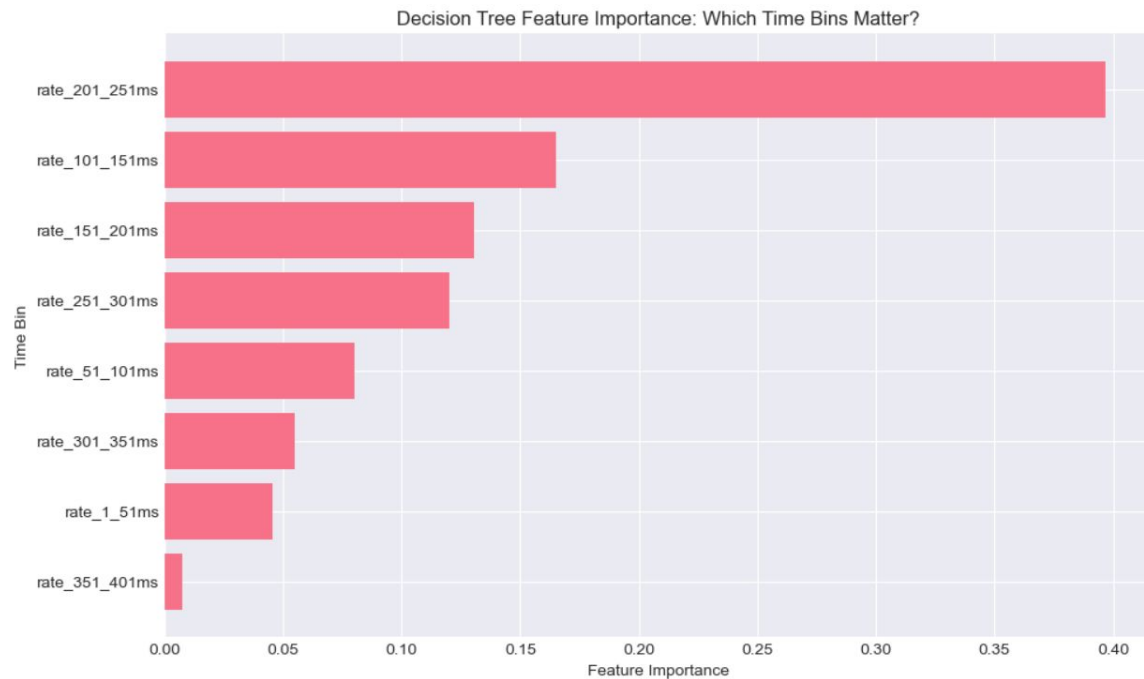


Methods Overview II

- Preprocessing: Still used mean firing rates, but preserved the rows from before without aggregation.
- Created a baseline decision tree. Pruned the tree after that to improve accuracy.

Key Findings II

The accuracy for the baseline was bad. The pruning resulted in a max_depth of 5. Accuracy only improved by 0.5%, though computational efficiency increased. Most important time bin was 200-250ms. This shows that we do not have enough proof that the data can be decoded to identify poses, further confirming pose-invariance. We can attempt to use other models like NNs and RF, at the loss of interpretability.



Decision Tree Visualization

